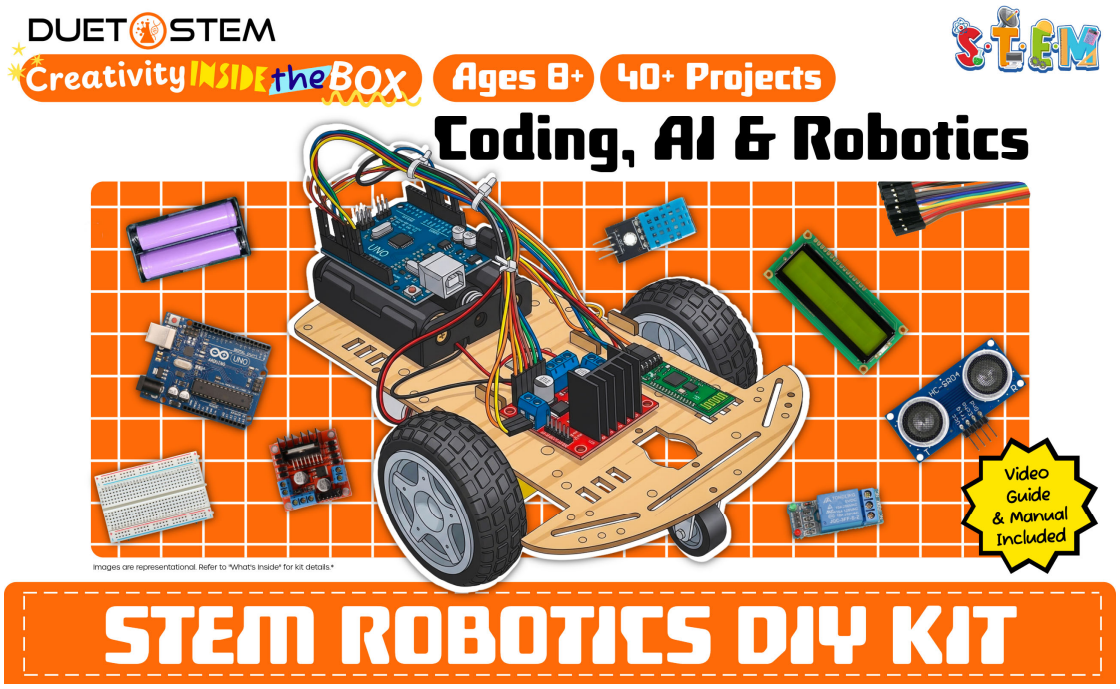
DUET STEM

PRODUCT & CURRICULUM DOSSIER

CODING, AI & ROBOTICS

STEM ROBOTICS DIY KIT

Creativity Inside the Box | Ages 8+ | 40+ Core Builds | 120+ Guided Projects



CLASSES	CLASS-WISE PROJECTS	ROBOTICS & INNOVATION LABS	CURRICULUM
3 – 12	100	20	NEP 2020 Aligned

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Prepared for: _____ Date: _____

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At a glance: one kit, ten classes, 120 documented projects — 100 mapped class by class from Class 3 to Class 12, plus 10 core robotics labs and 10 innovation, IoT and AI labs.

01 ABOUT DUET STEM

DUET STEM™ is the education technology brand of Duet Power Pvt. Ltd., Noida. We design and deliver complete Coding, AI and Robotics learning solutions for schools — hardware kits, structured curriculum, teacher training, digital resources and certification, supported by on-site trainers.

This document presents the DUET STEM Coding, AI & Robotics STEM Robotics DIY Kit: what the box contains, what a student can build with it, and exactly how those builds map to each class from Class 3 to Class 12.

Why this kit is different

- One kit, ten years of learning. The same box supports a Class 3 first circuit and a Class 12 AI capstone — schools do not need a different kit for every grade.
- A defined project for every class. 100 class-mapped projects plus 20 dedicated lab activities, each with a stated learning outcome and component list.
- Real robot, not a toy. A working 2WD robot chassis with geared motors, motor driver and caster wheel is included — students build a machine that actually drives.
- Sensor breadth. Six different sensor types cover distance, climate, light, flame, infrared and soil, so projects span science, environment, safety and agriculture.
- Path to AI. The Bluetooth module and serial interface let senior students connect the kit to Python, OpenCV and machine-learning workflows.
- Teacher-ready. Illustrated manual, video guides and trainer support mean a non-specialist teacher can run the lab from day one.

02 THE KIT AT A GLANCE

Product name	DUET STEM Coding, AI & Robotics — STEM Robotics DIY Kit
Recommended age	8 years and above
Class coverage	Class 3 to Class 12 (primary, middle and senior secondary)
Controller	Arduino Uno R3 (ATmega328P)
Robot platform	2WD laser-cut MDF chassis with caster wheel, geared BO motors and L298N driver
Sensors included	Ultrasonic, DHT11 temperature & humidity, LDR, flame, IR, soil moisture
Actuators included	4 x BO geared motors, 2 x servo motors
Output devices	I2C 16x2 LCD, 2 x buzzer, LED pack, relay module
Wireless	HC-05 Bluetooth module (smartphone and Python control)
Power	Rechargeable cells with holder and charger, plus a small solar panel
Programming	Block-based (Scratch-style) for juniors; Arduino C/C++ for middle and senior; Python integration for Classes 11–12
Core builds	40+ documented projects in the manual
Total project library	120 projects — 100 class-wise plus 20 robotics and innovation labs
Resources	Illustrated project manual, QR-linked video guides, circuit diagrams and sample code
Compliance	RoHS compliant | Made in India | For educational and learning purposes

Images are representational. Refer to the component list in Section 3 for the definitive kit contents.

03 WHAT'S INSIDE THE BOX

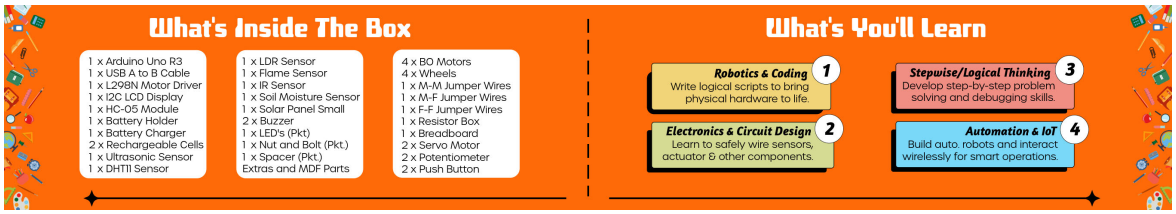
The kit ships as a single boxed set containing every component listed below, along with the printed manual and video-guide access card. Components are grouped by function for easy inventory and lab management.

Component	Qty	Purpose & Typical Use
A. Controller, Programming & Connectivity		
Arduino Uno R3 (ATmega328P)	1	The brain of every project — reads all sensors and drives all outputs. 14 digital + 6 analog pins.
USB A-to-B Programming Cable	1	Uploads code from laptop to the board and powers it during desk-based experiments.
HC-05 Bluetooth Module	1	Wireless serial link — enables app-controlled cars, voice commands and phone telemetry.
B. Mechanical Platform & Chassis		
2WD Car Chassis (laser-cut MDF)	1 set	Ready-to-assemble robot body with mounting slots for board, driver, battery and sensors.
Caster Wheel (free-rotating)	1	Third point of contact that balances the 2WD platform and allows smooth turning.
BO Geared Motors	4	Drive the robot. Two are used for the 2WD build; the spare pair enables a 4WD upgrade or replacements.
Rubber-Tyre Wheels	4	High-grip wheels matched to the BO motor shaft for reliable traction on tile and chart paper.
Nut & Bolt Assortment (packet)	1	Fastening hardware for chassis, motor brackets, sensor mounts and servo horns.
Spacer Set (packet)	1	Creates clearance between chassis decks and protects the PCB underside from shorting.
Extra MDF Parts & Accessories	1 set	Additional laser-cut brackets, arms, gripper plates and mounts for arm and turret builds.
C. Power System		
L298N Dual H-Bridge Motor Driver	1	Controls direction and speed (PWM) of two motors independently — essential for all mobile robots.
Battery Holder	1	Holds the rechargeable cells and supplies motor-grade current to the driver board.
Rechargeable Li-ion Cells	2	Portable power so the robot can run untethered from the laptop.
Battery Charger	1	Safely recharges the cells between classroom sessions.
Solar Panel (small)	1	Renewable-energy experiments: solar tracking, light-to-power conversion and energy logging.
D. Actuators & Motion		
Servo Motors (180°)	2	Precise angular control for robotic arms, grippers, sonar turrets, gates and smart-bin lids.
E. Sensors — Perception Layer		
Ultrasonic Sensor HC-SR04	1	Measures distance by sound echo. Used in obstacle avoidance, parking alarms, level sensing, radar.
DHT11 Temperature & Humidity Sensor	1	Digital climate sensing for weather stations, thermostats and greenhouse control.

Component	Qty	Purpose & Typical Use
LDR Light Sensor	1	Detects ambient light for night lamps, street lighting and solar-tracking projects.
Flame Sensor (IR)	1	Detects fire/flame signature for alarm systems and fire-fighting robots.
IR Sensor Module	1	Line detection, object counting, touch-free triggering and edge (cliff) detection.
Soil Moisture Sensor	1	Agri-tech projects — smart irrigation, plant health monitoring and data logging.
F. Output, Display & Switching		
I2C LCD Display (16x2)	1	Two-wire display for readings, menus and scoreboards without consuming many pins.
Buzzer	2	Audio alerts, melodies, Morse code and proximity warnings.
Assorted LEDs (packet)	1	Visual output for traffic lights, indicators, status displays and PWM experiments.
Relay Module	1	Lets a 5V Arduino switch a real appliance-style load — the gateway to home automation.
G. Prototyping, Wiring & Passives		
Solderless Breadboard	1	Build and rebuild circuits safely with no soldering — the core of every electronics lesson.
Male-to-Male Jumper Wires	1 set	Breadboard-to-breadboard and board-to-breadboard connections.
Male-to-Female Jumper Wires	1 set	Connects sensor module headers to the Arduino.
Female-to-Female Jumper Wires	1 set	Module-to-module connections such as I2C LCD and HC-05.
Resistor Box (assorted values)	1	Current limiting, pull-up/pull-down networks and Ohm's law experiments.
Potentiometers	2	Analog input for dimmers, speed dials and master-slave robotic arm control.
Push Buttons	2	Digital input for triggers, keypads, reaction games and password locks.
H. Learning Resources (included)		
Illustrated Project Manual	1	Step-by-step circuit diagrams, code listings and challenge extensions.
Video Guide Access (QR-linked)	1	Build-along videos for every core project, usable by both students and teachers.

Notes on quantities

- Four BO motors and four wheels are supplied against a 2WD build. The additional pair gives the school a 4WD upgrade option and a spare set for the life of the kit.
- The chassis is supplied as flat laser-cut MDF parts with a caster wheel, nuts, bolts and spacers — assembly is itself a designed learning activity.
- Three jumper-wire types (M-M, M-F, F-F) are included so every module and breadboard combination can be wired without adapters.



Packaging artwork: contents and learning outcomes panel.

04 LEARNING OUTCOMES & SKILL PROGRESSION

The project library is deliberately sequenced. Each stage assumes the previous one and adds a new class of capability, so a school running the kit from Class 3 upwards never repeats content.

Stage	Classes	What the student can do by the end of the stage
Foundation	**Class 3 – 5**	Identify components, build safe circuits on a breadboard, understand input–process–output, upload and edit basic Arduino sketches, read a sensor and act on it.
Builder	**Class 6 – 8**	Assemble and program a working robot, combine multiple sensors, control motors and servos with PWM, use wireless control and switch real loads through a relay.
Engineer	**Class 9 – 10**	Apply feedback and proportional control, log and graph data, implement navigation algorithms, and take a problem statement through to a demonstrable product.
Innovator	**Class 11 – 12**	Bridge hardware to Python, build datasets, apply machine learning and computer vision, and deliver a documented autonomous system as a capstone.

Skills developed across the programme

Technical skills

- Circuit building and breadboard prototyping
- Embedded programming in Arduino C/C++
- Sensor interfacing and calibration
- Motor control, PWM and H-bridge drive
- Wireless communication over Bluetooth
- Python, data logging and machine learning

21st-century skills

- Computational and stepwise logical thinking
- Debugging, persistence and iteration
- Design thinking and problem framing
- Collaboration in build teams
- Scientific documentation and presentation
- Innovation and entrepreneurial mindset

05 CURRICULUM ALIGNMENT

Framework	How this kit maps to it
NEP 2020	Supports experiential, hands-on and multidisciplinary learning, and the introduction of coding from Class 6 onwards. Vocational and skill exposure is embedded from the primary stage.
CBSE	Directly supports Artificial Intelligence (Code 417) and Information Technology at secondary level, and provides ready-made investigatory and science-exhibition projects for Classes 9–12.
ICSE / ISC & State Boards	Covers practical components of Computer Applications, Physics (electricity, energy, sound) and Environmental Studies through applied builds.
ATL / NITI Aayog	Matches the tinkering, prototyping and innovation objectives of Atal Tinkering Labs; the kit can be deployed as a core ATL resource.
Science & Maths integration	Ohm's law, series and parallel circuits, speed of sound, energy conversion, data handling, averages and graphing are all taught through working hardware.

06 PROJECT LIBRARY — OVERVIEW

Every project below is buildable with the components supplied in this single kit. No additional purchase is required for any class-wise project except where a project explicitly uses a laptop for Python and computer-vision work in Classes 11 and 12.

Class	Theme	Projects	Programming Mode
Class 3	Discover — Circuits, Motion & Making	**10**	Guided / no-code, first upload
Class 4	Control — First Code, First Commands	**10**	Block-based & basic Arduino C
Class 5	Sense — Reading the World	**10**	Arduino C, Serial Monitor, LCD
Class 6	Build — Your First Robots	**10**	Arduino C, motor & servo control
Class 7	Integrate — Multi-Sensor Machines	**10**	Arduino C, multi-sensor logic
Class 8	Automate — Smart Systems	**10**	Arduino C, relay, state machines
Class 9	Engineer — Control, Data & Algorithms	**10**	Arduino C, control loops, plotting
Class 10	Innovate — Real-World Problem Solving	**10**	Arduino C, full system integration
Class 11	Apply AI — Data, Python & ML	**10**	Arduino C + Python, scikit-learn
Class 12	Capstone — Autonomous & Intelligent Systems	**10**	Python, OpenCV, ML deployment
Sub-total		**100**	**Class-mapped projects**
Robotics Labs		**10**	Core robotics skill builds
Innovation Labs		**10**	Electronics, IoT, energy & AI
TOTAL PROJECT LIBRARY		**120**	**From one kit**

07 CLASS-WISE PROJECT PLAN

Ten projects are specified for each class from Class 3 to Class 12. Each project states the build, the learning outcome and the components involved, so a teacher can plan a full academic year directly from this table.

Class 3 — Discover — Circuits, Motion & Making

No-code and first-code exploration. Students handle components, complete circuits and understand cause and effect.

#	Project	What the student builds and learns	Key components used
1	**Light the Lamp**	Build a first closed circuit and discover why a break in the loop stops the glow.	*LED, resistor, breadboard, jumper wires, battery holder*
2	**Press-to-Buzz Doorbell**	Wire a push button so the buzzer sounds only while pressed — input causes output.	*Push button, buzzer, breadboard*
3	**Traffic Light Board**	Lay out red, yellow and green LEDs and switch them by hand to act out road-safety rules.	*3 LEDs, resistors, breadboard*
4	**Colour-Code Detective**	Read resistor colour bands and sort the resistor box into a value chart.	*Resistor box*
5	**Dimmer Knob**	Turn a potentiometer and watch an LED fade — the first feel for 'more or less' instead of 'on or off'.	*Potentiometer, LED, resistor*
6	**Torch & Sensor Game**	Shine a torch on the LDR and see the light change; introduces the idea of a sensor.	*LDR, LED, breadboard*
7	**Spin the Wheel**	Connect a BO motor to the battery and fit a wheel; observe the direction reverse when wires swap.	*BO motor, wheel, battery holder*
8	**Sun Power Fan**	Run a motor directly off the solar panel and hunt for the brightest spot in the room.	*Solar panel, BO motor*
9	**Nuts, Bolts & Chassis**	Guided mechanical build of the 2WD chassis using spacers, nuts and bolts.	*MDF chassis, caster wheel, nuts, bolts, spacers*
10	**Hello, Blink!**	Upload the very first Arduino sketch and make the on-board LED blink.	*Arduino Uno R3, USB cable*

Extension challenges

- Redesign the traffic-light board for a four-way junction using extra LEDs.
- Decorate and name your robot chassis, then present the build to the class.
- Find three objects in school that use a sensor and explain what each one senses.

Assessment checkpoint

Can the student build a working circuit from a diagram and explain what each part does?

Projects in this class: 10

Class 4 — Control — First Code, First Commands

Students write and modify their own sketches: digital output, digital input, tone and PWM.

#	Project	What the student builds and learns	Key components used
1	**Blink Your Own LED**	Move blink to an external LED on a digital pin and change the timing values.	*Arduino, LED, resistor*
2	**SOS Signal Lamp**	Program a Morse-code distress light using long and short pulses.	*Arduino, LED*
3	**Alternating Beacon**	Two LEDs flashing in opposite phase, like an ambulance light bar.	*Arduino, 2 LEDs, resistors*
4	**Beep Machine**	Play tones and a simple tune on the buzzer using the tone() function.	*Arduino, buzzer*
5	**Button-Controlled Light**	Read a push button with digitalRead() and toggle an LED on and off.	*Arduino, push button, LED*
6	**Musical Doorbell**	Combine a button with a melody; introduces arrays and custom functions.	*Arduino, button, buzzer*
7	**Automatic Night Lamp**	The LDR switches the LED on when the room gets dark — first sensor-driven decision.	*Arduino, LDR, LED*
8	**Brightness Dial**	Potentiometer controls LED brightness through analogWrite() and PWM.	*Arduino, potentiometer, LED*
9	**Servo Pointer Gauge**	Sweep a servo, then steer it live with the potentiometer knob.	*Arduino, servo, potentiometer*
10	**Motor Driver First Run**	Drive a BO motor forward, backward and stop using the L298N driver.	*Arduino, L298N, BO motor, battery pack*

Extension challenges

- Change the SOS lamp to flash your own initials in Morse code.
- Make the night lamp switch on only when it is dark AND the button is pressed.
- Program the servo to point at a number you choose on a paper dial.

Assessment checkpoint

Can the student modify supplied code to change behaviour and explain the change?

Projects in this class: 10

Class 5 — Sense — Reading the World

Analog and digital sensor reading, threshold logic, serial monitor and the I2C LCD.

#	Project	What the student builds and learns	Key components used
1	**Smart Traffic Light**	Full traffic sequence with a pedestrian request button and walk buzzer.	*Arduino, 3 LEDs, push button, buzzer*
2	**Electronic Dice**	Generate a random number on button press and show it as an LED pattern.	*Arduino, LEDs, push button*
3	**Distance Meter**	Print live ultrasonic readings in centimetres to the Serial Monitor.	*Arduino, HC-SR04*
4	**Reverse Parking Alarm**	Beep faster as an obstacle gets nearer — a real automotive feature, modelled.	*Arduino, HC-SR04, buzzer*
5	**Room Climate Display**	Show live temperature and humidity on the I2C LCD.	*Arduino, DHT11, I2C LCD*
6	**Fever Alert Thermometer**	Trigger an LED and buzzer when temperature crosses a set threshold.	*Arduino, DHT11, LED, buzzer*
7	**Black or White?**	The IR sensor distinguishes a black line from the floor; result shown on the LCD.	*Arduino, IR sensor, LCD*
8	**Smart Blind Stick (model)**	A proximity warning device that beeps when something is close ahead.	*Arduino, HC-SR04, buzzer*
9	**Thirsty Plant Indicator**	The soil moisture sensor lights a 'water me' LED when the pot dries out.	*Arduino, soil moisture sensor, LED*
10	**Solar Light Meter**	Measure and compare solar panel output under different lighting conditions.	*Solar panel, LDR, Arduino, LCD*

Extension challenges

- Make the parking alarm change tone, not just speed, as distance shrinks.
- Display both the temperature and a plain-language 'comfort' message on the LCD.
- Record ten soil-moisture readings across a week and plot them on graph paper.

Assessment checkpoint

Can the student read a sensor and write correct threshold logic without help?

Projects in this class: 10

Class 6 — Build — Your First Robots

The chassis comes alive. Students combine sensing, decision-making and motion into autonomous machines.

#	Project	What the student builds and learns	Key components used
1	**Obstacle-Avoiding Robot**	The robot detects walls ahead and steers around them without any human input.	*Chassis, HC-SR04, L298N, BO motors*
2	**Line-Following Robot**	Single-sensor edge-following bot that tracks a black line drawn on chart paper.	*IR sensor, chassis, L298N, motors*
3	**Bluetooth RC Car**	Drive the robot from a smartphone app over the HC-05 wireless link.	*HC-05, chassis, L298N, motors*
4	**Smart Dustbin**	A servo-driven lid that opens automatically when a hand approaches.	*HC-SR04, servo*
5	**Automatic Street Light**	The LDR switches a lamp through the relay at dusk and off at dawn.	*LDR, relay module, LED*
6	**Fire Alarm System**	The flame sensor triggers an instant buzzer and warning light.	*Flame sensor, buzzer, LED*
7	**Temperature-Controlled Fan**	A BO-motor fan starts automatically above a set temperature; status on LCD.	*DHT11, BO motor, L298N, LCD*
8	**Sonar Scanner Turret**	A servo sweeps the ultrasonic sensor and plots distances across a 180° arc.	*Servo, HC-SR04*
9	**Plant Care Station**	Combined soil, temperature and humidity monitoring with LCD alerts.	*Soil moisture, DHT11, LCD, buzzer*
10	**Speed-by-Distance Robot**	The closer the obstacle, the slower the motors run — proportional PWM control.	*HC-SR04, L298N, motors*

Extension challenges

- Make the obstacle-avoiding robot choose left or right based on which side is clearer.
- Race two line followers and tune yours for the fastest clean lap.
- Add a reversing buzzer and an indicator LED to your robot.

Assessment checkpoint

Can the student build, wire and program a working autonomous robot end to end?

Projects in this class: 10

Class 7 — Integrate — Multi-Sensor Machines

Two or more sensors working together, wireless control and the first mechanisms with feedback.

#	Project	What the student builds and learns	Key components used
1	**Voice-Controlled Robot Car**	Speak a command to the phone app and the robot executes it over Bluetooth.	*HC-05, chassis, L298N*
2	**Hand-Follower Robot**	The robot maintains a fixed distance from your palm, advancing and retreating.	*HC-SR04, chassis, motors*
3	**Servo Door Lock**	A button-sequence password unlocks a servo-driven latch; attempts shown on LCD.	*Servo, push buttons, LCD*
4	**Touch-Free Dispenser**	IR detection triggers a motorised dispensing action with no contact.	*IR sensor, servo, BO motor*
5	**Mini Weather Station**	Temperature, humidity and light level presented together on a single screen.	*DHT11, LDR, I2C LCD*
6	**Fire-Fighting Robot**	The robot detects a flame, drives towards it and activates a fan to extinguish it.	*Flame sensor, chassis, BO motor fan*
7	**Sun-Tracking Solar Panel**	A sweep-and-seek algorithm rotates the panel to the brightest angle.	*LDR, servo, solar panel*
8	**Tuned Line Follower**	Add PWM speed control and tune the turn response for faster, smoother laps.	*IR sensor, L298N, chassis*
9	**Reaction-Time Game**	Two-player buzzer game with an LCD scoreboard and random start delay.	*Push buttons, LCD, buzzer*
10	**Anti-Theft Bag Alarm**	An IR trip triggers a loud alert when the bag is opened without permission.	*IR sensor, buzzer, Arduino*

Extension challenges

- Add an LCD to the Bluetooth car that shows the last command received.
- Make the sun tracker record its best angle every hour of the day.
- Add a wrong-password lockout with an alarm to the servo door lock.

Assessment checkpoint

Can the student combine two or more sensors into a single, reliable decision?

Projects in this class: 10

Class 8 — Automate — Smart Systems & Home Automation

Relay switching, state machines, dual-mode control and the first two-axis mechanism.

#	Project	What the student builds and learns	Key components used
1	**Dual-Mode Robot**	One robot, two modes: fully autonomous obstacle avoidance or manual Bluetooth drive.	*HC-05, HC-SR04, chassis, L298N*
2	**Smart Irrigation System**	Soil moisture drops below threshold, the relay starts the pump, LCD logs the event.	*Soil moisture, relay, LCD*
3	**Bluetooth Home Automation**	Switch an appliance-style load on and off from a phone using the relay.	*HC-05, relay module*
4	**Water Tank Level Indicator**	Ultrasonic level sensing with LED bar graph and overflow alarm.	*HC-SR04, LEDs, buzzer, LCD*
5	**Fire Safety with Auto Cut-Off**	On flame detection the system alarms and cuts the load through the relay.	*Flame sensor, relay, buzzer*
6	**2-Axis Robotic Arm**	Build a base-and-gripper arm and program a repeatable pick-and-place routine.	*2 servos, MDF parts, Arduino*
7	**Bluetooth Notice Board**	Send any message from a phone and display it live on the I2C LCD.	*HC-05, I2C LCD*
8	**Automatic Parking Gate**	IR detection raises a servo barrier and logs vehicle count on the LCD.	*IR sensor, servo, LCD*
9	**Thermostat with Hysteresis**	Prevent relay chatter by adding a dead band around the set point.	*DHT11, relay, LCD*
10	**Solar Energy Logger**	Track panel output through the day and compare orientations.	*Solar panel, Arduino, LCD*

Extension challenges

- Add a manual override button to the smart irrigation system.
- Program the robotic arm to sort two different objects into separate bins.
- Extend home automation to two loads with confirmed status feedback on the phone.

Assessment checkpoint

Can the student design a system with clear states, thresholds and safe defaults?

Projects in this class: 10

Class 9 — Engineer — Control, Data & Algorithms

Closed-loop control, data capture and graphing, algorithmic navigation and measurement science.

#	Project	What the student builds and learns	Key components used
1	**Serial Data Logging & Graphing**	Stream DHT11 readings and visualise them live on the Serial Plotter.	*Arduino, DHT11, Serial Plotter*
2	**Proportional Line Follower**	Replace on/off steering with proportional control for smooth, fast tracking.	*IR sensor, L298N, chassis*
3	**Sonar Mapping**	A servo-swept ultrasonic builds a distance map of the surrounding space.	*Servo, HC-SR04*
4	**Telemetry Robot**	The robot streams live sensor data back to the phone while it drives.	*HC-05, sensors, chassis*
5	**Smart Greenhouse**	Soil, temperature and humidity together drive an automated pump and fan.	*Soil moisture, DHT11, relay, motor*
6	**Wall-Following Maze Robot**	Implement the left-hand rule to navigate a cardboard maze autonomously.	*HC-SR04, chassis, L298N*
7	**Master–Slave Robotic Arm**	Two potentiometers act as a control console for the two servo joints.	*2 potentiometers, 2 servos*
8	**Smart Street-Lighting Grid**	A solar-charged, LDR-triggered lighting model with relay switching.	*Solar panel, LDR, relay, LEDs*
9	**Conveyor Object Counter**	A motor-driven belt with IR counting and a running total on the LCD.	*IR sensor, LCD, BO motor*
10	**Speed of Sound Experiment**	Use known distances and echo timings to calculate the speed of sound.	*HC-SR04, Arduino*

Extension challenges

- Tune the proportional gain and tabulate lap time against gain value.
- Plot the sonar map on graph paper and identify the widest available opening.
- Compare your measured speed of sound with the textbook value and explain the error.

Assessment checkpoint

Can the student collect data, analyse it and justify a design decision from it?

Projects in this class: 10

Class 10 — Innovate — Real-World Problem Solving

Board-project standard builds: complete products with a defined user, problem statement and demonstration.

#	Project	What the student builds and learns	Key components used
1	**Autonomous Delivery Rover**	A rover that follows a route, avoids obstacles and stops at a marked destination.	*Chassis, HC-SR04, IR, L298N*
2	**Agri-Bot Soil Surveyor**	A mobile rover that samples soil moisture at multiple points and reports wirelessly.	*Soil moisture, chassis, HC-05*
3	**Autonomous Fire-Fighting Robot**	Flame seeking, approach, extinguish and retreat — fully autonomous sequence.	*Flame sensor, chassis, BO motor fan*
4	**Smart Assistive Cane**	Ultrasonic obstacle detection with layered audio and vibration feedback.	*HC-SR04, buzzer, BO motor*
5	**Anti-Collision Braking Model**	An ADAS-style demonstrator that brakes automatically before impact.	*HC-SR04, L298N, chassis*
6	**Solar Tracker with Logging**	Dual-axis-style tracking with recorded energy gain versus a fixed panel.	*Servo, LDR, solar panel*
7	**Smart Waste Bin**	Automatic lid, fill-level measurement and a wireless 'bin full' alert.	*HC-SR04, servo, HC-05*
8	**Home Automation Hub**	Multi-device control with status feedback shown on both phone and LCD.	*HC-05, relay, LCD*
9	**Weather Station with Trends**	Record readings over hours and present maxima, minima and averages.	*DHT11, LDR, LCD*
10	**Smart Traffic Junction**	Science-fair capstone: adaptive signal timing based on vehicle detection.	*LEDs, IR sensor, servo, LCD*

Extension challenges

- Write a one-page problem statement, user profile and solution brief.
- Add fail-safe behaviour: decide what the robot does if a sensor stops responding.
- Prepare a three-minute demonstration and an exhibition poster for the project.

Assessment checkpoint

Can the student take a real problem through to a working, documented product?

Projects in this class: 10

Class 11 — Apply AI — Data, Python & Machine Learning

The kit becomes a data source. Students bridge Arduino to Python, build datasets and train their first models.

#	Project	What the student builds and learns	Key components used
1	**Arduino to Python Bridge**	Stream sensor data over serial into Python and plot a live dashboard.	*Arduino, PySerial, Matplotlib*
2	**Build Your Own Dataset**	Log DHT11, LDR and soil readings to CSV and clean the data for analysis.	*All sensors, Python, Pandas*
3	**Gesture-Controlled Robot**	Train a webcam gesture model and use it to drive the robot over Bluetooth.	*HC-05, chassis, webcam model*
4	**Face-Following Robot**	Computer vision on the laptop steers the robot to keep a face centred.	*HC-05, chassis, OpenCV*
5	**Voice-Command Robot**	Python speech recognition converts spoken words into motion commands.	*HC-05, chassis, Python*
6	**Temperature Trend Prediction**	Fit a regression model to logged climate data and forecast the next reading.	*DHT11 dataset, scikit-learn*
7	**Soil Anomaly Detection**	Spot irregular irrigation events in a moisture time series using ML.	*Soil sensor, Python*
8	**Light-Based Object Sorter**	An LDR classifies objects by reflectivity and a servo arm sorts them.	*LDR, servo, BO motor*
9	**Footfall Counter & Analytics**	IR-based people counting with hourly analysis and a plotted report.	*IR sensor, Python*
10	**Rules vs Learning**	Compare a hand-tuned line follower against thresholds derived from logged data.	*IR sensor, chassis, Python*

Extension challenges

- Extend your dataset to 500+ rows and report the accuracy of the model.
- Add a confidence threshold so the robot ignores uncertain predictions.
- Replace one hard-coded threshold from an earlier project with a learned value.

Assessment checkpoint

Can the student build a dataset, train a model and connect it back to hardware?

Projects in this class: 10

Class 12 — Capstone — Autonomous & Intelligent Systems

Full-system, portfolio-grade projects suitable for board investigatory submissions, hackathons and exhibitions.

#	Project	What the student builds and learns	Key components used
1	**AI Vision-Guided Rover**	Object and colour tracking on the laptop guides an autonomous ground vehicle.	*OpenCV, HC-05, chassis*
2	**Integrated Smart City Model**	Traffic, street lighting, parking and irrigation running as one connected system.	*Full kit*
3	**Hand-Tracked Robotic Arm**	Live hand landmarks from a webcam map directly onto the two servo joints.	*2 servos, webcam, Python*
4	**Predictive Irrigation with ML**	A trained model decides when to water, not a fixed threshold.	*Soil, DHT11, relay, Python*
5	**Autonomous Maze Solver**	Explore, remember and then run the shortest path through an unknown maze.	*HC-SR04, chassis, L298N*
6	**Speech-to-Action Smart Home**	Natural spoken commands operate relay-controlled loads with confirmation.	*HC-05, relay, Python*
7	**Renewable Energy Study**	Quantify solar output against angle and irradiance and report the efficiency curve.	*Solar panel, LDR, servo*
8	**Bluetooth IoT Dashboard**	A local web dashboard showing live kit sensor data with historical charts.	*HC-05, Python, Flask*
9	**Assistive Technology Capstone**	A smart mobility aid prototype with obstacle detection and guided steering.	*Chassis, HC-SR04, servo*
10	**Investigatory Project & Viva**	Complete write-up, circuit documentation, code repository and demonstration.	*Full kit*

Extension challenges

- Produce full documentation: abstract, circuit, code, results and conclusion.
- Benchmark the autonomous system over ten runs and report the success rate.
- Identify one limitation of the prototype and propose a costed improvement.

Assessment checkpoint

Can the student deliver, demonstrate and defend a complete autonomous system?

Projects in this class: 10

08 10 CORE ROBOTICS LAB ACTIVITIES

These ten activities form the robotics backbone of the programme. They can be run as a dedicated robotics club module, a competition-preparation track, or as reinforcement alongside the class-wise plan.

#	Activity	Description	Skills developed
1	**Chassis Assembly & Mechanical Design**	Build the 2WD platform: motor mounting, caster alignment, weight balance and cable routing.	*Mechanical design, tool handling, spatial reasoning*
2	**Differential Drive & Motor Calibration**	Make the robot travel straight, turn on the spot and stop accurately by trimming PWM per motor.	*Kinematics, calibration, systematic testing*
3	**Obstacle-Avoiding Robot**	Continuous ultrasonic scanning with a decision tree for stop, reverse and turn.	*Conditional logic, sensor fusion, autonomy*
4	**Line-Following Robot**	Track a printed path using IR reflectance, with tuning for speed and curve radius.	*Feedback control, tuning, iteration*
5	**Edge & Cliff Detection**	Table-top safety behaviour — the robot detects a missing surface and retreats.	*Safety design, fail-safe logic*
6	**Bluetooth Remote-Controlled Robot**	Full manual drive from a phone app with a defined command protocol.	*Wireless communication, protocol design*
7	**Wall-Following & Maze Navigation**	Implement the left-hand rule and measure completion time across maze layouts.	*Algorithms, pathfinding, benchmarking*
8	**Light-Seeking (Photovore) Robot**	The robot hunts the brightest region in the room using LDR feedback.	*Analog sensing, bio-inspired robotics*
9	**Servo Sonar Scanning Turret**	A 180° sweeping ultrasonic head that reports the clearest direction of travel.	*Polar mapping, servo control, data plotting*
10	**2-DOF Robotic Arm with Gripper**	Base rotation plus gripper actuation, programmed for repeatable pick-and-place.	*Manipulator control, sequencing, precision*

09 10 INNOVATION, ELECTRONICS, IOT & AI ACTIVITIES

Beyond mobile robots, the same kit supports a full electronics, energy, IoT and artificial-intelligence lab. These ten activities extend the programme into science, environment and data.

#	Activity	Description	Skills developed
1	**Breadboard & Ohm's Law Lab**	Build series and parallel circuits, measure and verify V = IR with the resistor box.	*Physics: current, voltage, resistance*
2	**Resistor Colour Code Workshop**	Decode, sort and verify resistor values; design a correct current-limiting network.	*Numeracy, standards, precision*
3	**Digital vs Analog & PWM Lab**	Compare digitalWrite and analogWrite; visualise duty cycle through LED brightness and motor speed.	*Signals, computer science fundamentals*
4	**Sensor Calibration & Error Analysis**	Map raw sensor values to real-world units and quantify drift and repeatability.	*Measurement science, statistics*
5	**Relay Home Automation Lab**	Safely switch a load from a microcontroller and design the control interface.	*Electrical safety, systems design*
6	**Smart Irrigation & Agri-Tech Lab**	Design a full watering control loop and justify the threshold you choose.	*Sustainability, applied engineering*
7	**Weather Station & Environment Lab**	Multi-parameter monitoring with LCD reporting and daily comparison.	*Environmental science, data literacy*
8	**Solar Energy & Renewables Lab**	Investigate how angle, distance and light intensity change panel output.	*Energy, experimental method*
9	**Data Logging, Graphing & Analysis**	Capture time-series data, export it and produce charts with written conclusions.	*Data handling, scientific writing*
10	**AI & Machine Learning Integration**	Connect the kit to Python for computer vision, speech and predictive models.	*AI/ML literacy, Python programming*

10 SUGGESTED DELIVERY & SESSION PLAN

The library is designed around a standard Indian school calendar of approximately 30 to 36 working weeks. One kit supports a group of two to three students in a shared lab, or a single student in a one-to-one programme.

Stage	Session length	Frequency	Coverage across the academic year
Class 3 – 5	40 minutes	1 period / week	10 projects per class, with repeat-and-extend challenges in the remaining weeks
Class 6 – 8	45 – 60 minutes	1 – 2 periods / week	10 projects per class plus 3 – 4 robotics lab activities
Class 9 – 10	60 minutes	2 periods / week	10 projects per class plus board investigatory project preparation
Class 11 – 12	60 – 90 minutes	2 periods / week	10 projects per class plus an individual or team capstone with documentation

Recommended lesson structure *(any project, any class)*

- Hook (5 min) — present the real-world problem the project solves.
- Circuit build (10–15 min) — students wire the components from the manual diagram.
- Code and upload (15 min) — type, upload and observe.
- Test and debug (10 min) — deliberate fault-finding; the highest-value part of the lesson.
- Challenge extension (5–10 min) — a stated modification that pushes beyond the manual.
- Record (5 min) — students log the outcome in their project journal.

11 DELIVERABLES, TRAINING & SUPPORT

Supplied with every kit

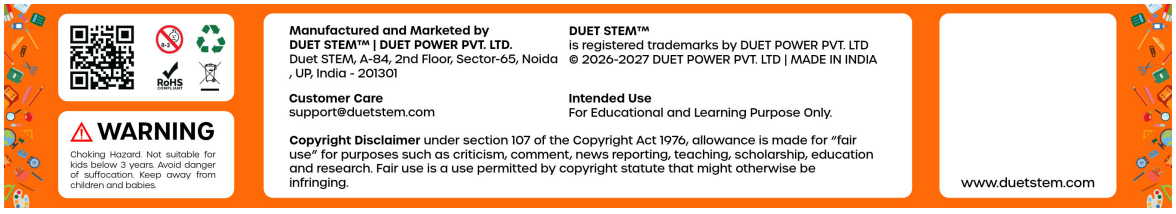
- Complete component set as per Section 3
- Assembled-ready 2WD chassis parts and hardware
- Illustrated step-by-step project manual
- QR-linked video build guides
- Circuit diagrams and ready sample code
- Challenge cards and project journal template

Available as programme add-ons

- Teacher training and certification workshops
- On-site dedicated trainer for regular classes
- Full Class 1–12 structured curriculum documents
- Student and teacher certification
- Complete AI & Robotics lab setup with furniture and branding
- Competition and exhibition mentoring

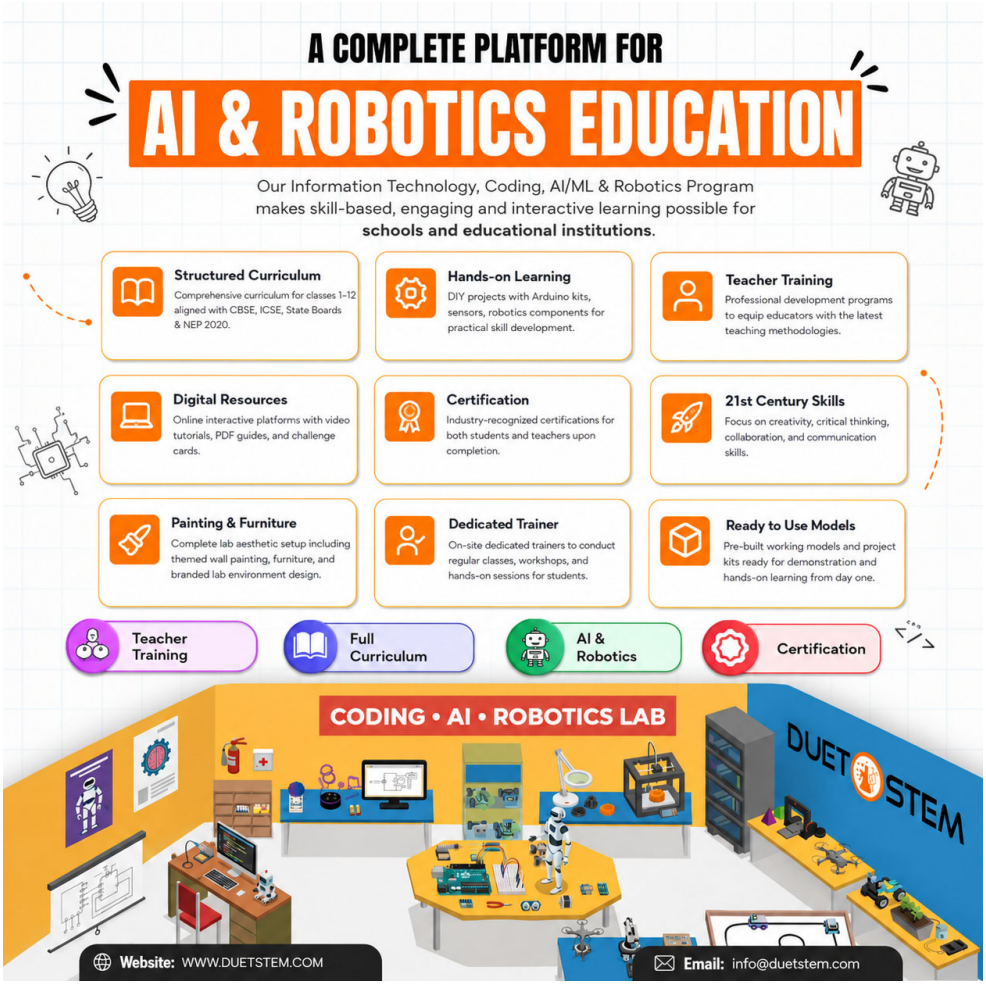
12 SAFETY, COMPLIANCE & WARRANTY

- Age guidance: recommended for ages 8 years and above under teacher or parental supervision.
- Choking hazard: the kit contains small parts. Not suitable for children below 3 years. Keep away from infants and toddlers.
- Electrical safety: all logic operates at 5V DC. Motors and loads run from the supplied rechargeable pack. The relay module must only be used with teacher supervision and with the loads specified in the manual.
- Battery care: use only the supplied charger. Do not short, puncture or expose cells to heat. Remove cells during long storage.
- Compliance: RoHS compliant. Manufactured and marketed in India by Duet Power Pvt. Ltd.
- Intended use: for educational and learning purposes only.
- Warranty: manufacturing defects covered as per the DUET STEM standard warranty policy. Consumables such as LEDs, jumper wires and cells are excluded.



13 THE DUET STEM LAB ECOSYSTEM

This kit is one component of a complete school programme. Where a school wishes to move beyond individual kits, DUET STEM designs, supplies and operates full laboratories.





Complete solutions available from DUET STEM

- ATL Labs — tinkering, prototyping, 3D printing and IoT, aligned with NITI Aayog ATL guidelines.
- STEM Labs — integrated Science, Technology, Engineering and Maths kits and experiments for all grade levels.
- AI & Robotics Labs — smart kits, sensors and controllers with a full coding and robotics progression.
- Science Labs — Physics, Chemistry, Biology and General Science with quality instruments and safety focus.
- AR & VR Labs — immersive simulations and 3D interactive content.
- Astronomy Labs — telescopes, space models and exploration kits.
- Lab painting, furniture, branding, teacher training, dedicated on-site trainers and certification.

14 COMMERCIAL SUMMARY & NEXT STEPS

Item	DUET STEM Coding, AI & Robotics — STEM Robotics DIY Kit
Unit of supply	1 boxed kit (complete as per Section 3)
Unit price	To be confirmed as per order quantity
Minimum order quantity	To be confirmed
Taxes	GST as applicable
Delivery	To be confirmed from date of confirmed purchase order
Payment terms	As per agreed commercial terms
Validity of offer	30 days from the date of this document

Next steps

- Confirm quantity and delivery locations so a formal quotation can be issued.
- Schedule a live demonstration of the kit and three sample projects at your campus.
- Select the add-on modules required: teacher training, on-site trainer, curriculum documents or full lab setup.

Creativity Inside the Box

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